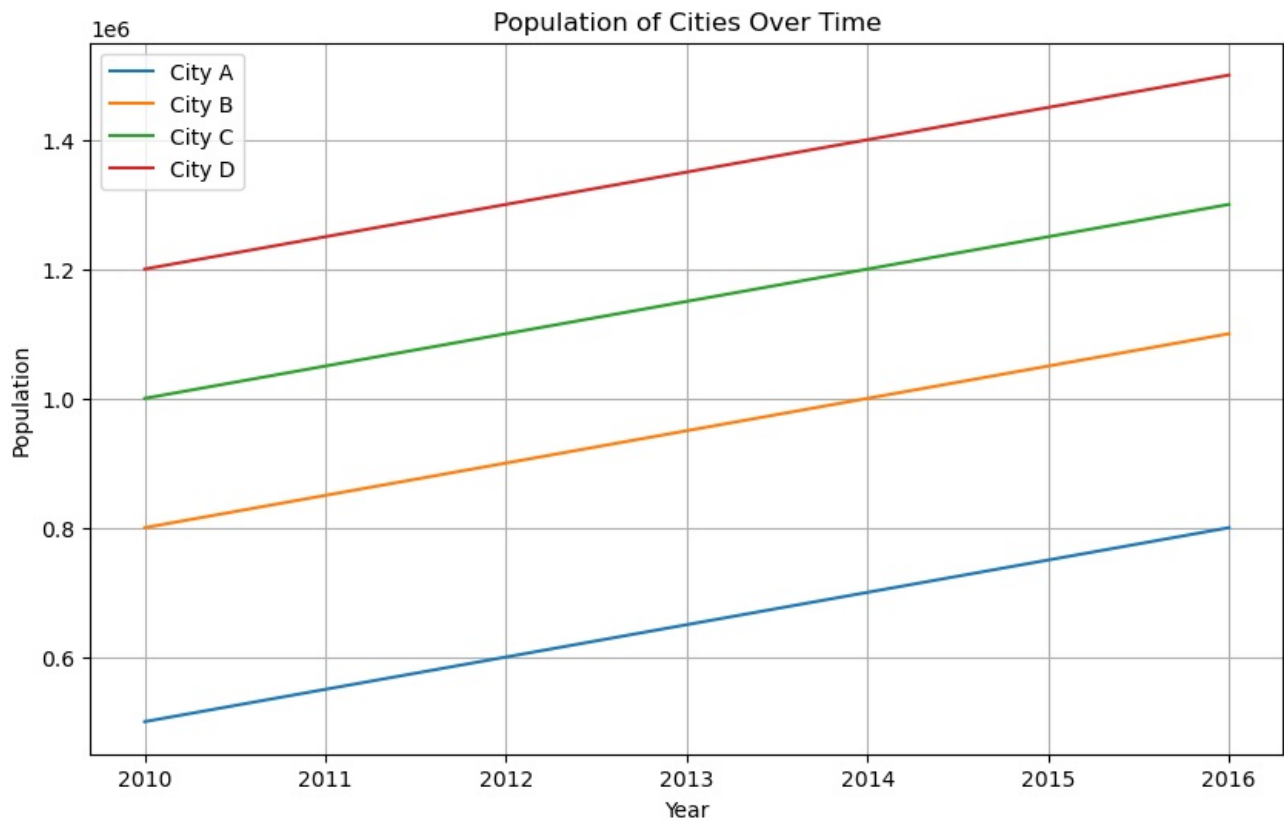


```
In [1]: import matplotlib.pyplot as plt
```

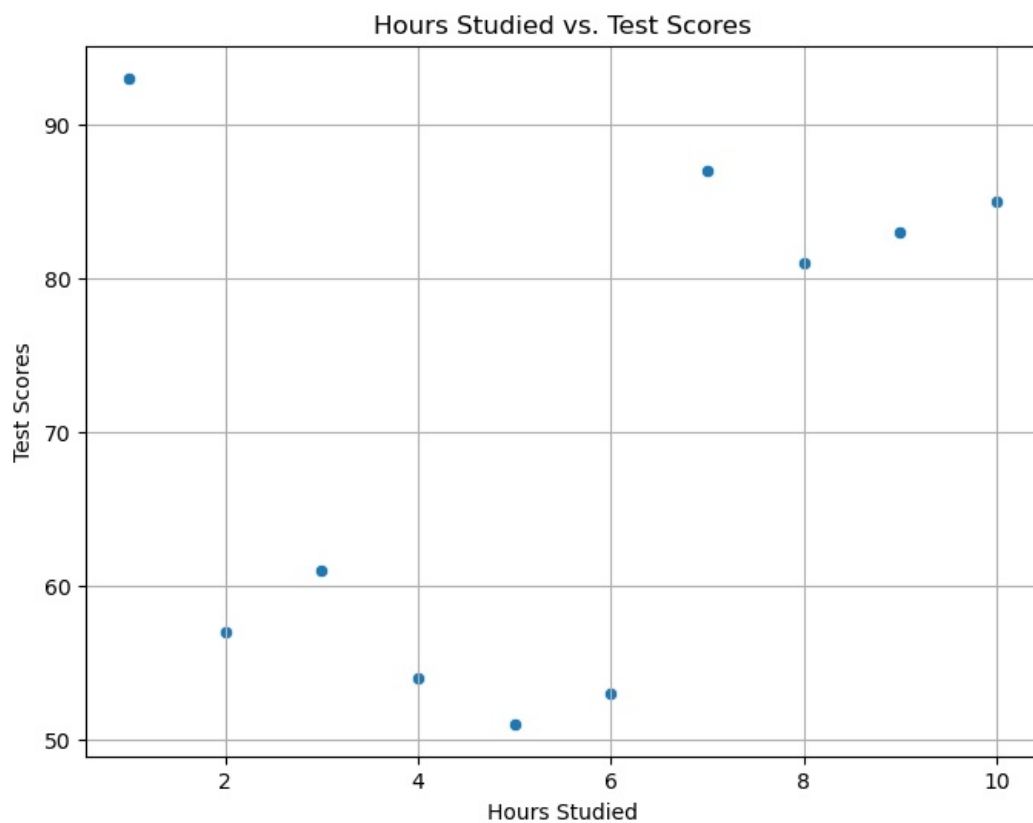
```
# Data for the four cities
years = [2010, 2011, 2012, 2013, 2014, 2015, 2016]
city_a = [500000, 550000, 600000, 650000, 700000, 750000, 800000]
city_b = [800000, 850000, 900000, 950000, 1000000, 1050000, 1100000]
city_c = [1000000, 1050000, 1100000, 1150000, 1200000, 1250000, 1300000]
city_d = [1200000, 1250000, 1300000, 1350000, 1400000, 1450000, 1500000]

# Create a line plot
plt.figure(figsize=(10, 6))
plt.plot(years, city_a, label='City A')
plt.plot(years, city_b, label='City B')
plt.plot(years, city_c, label='City C')
plt.plot(years, city_d, label='City D')
plt.title('Population of Cities Over Time')
plt.xlabel('Year')
plt.ylabel('Population')
plt.legend()
plt.grid(True)
plt.show()
```



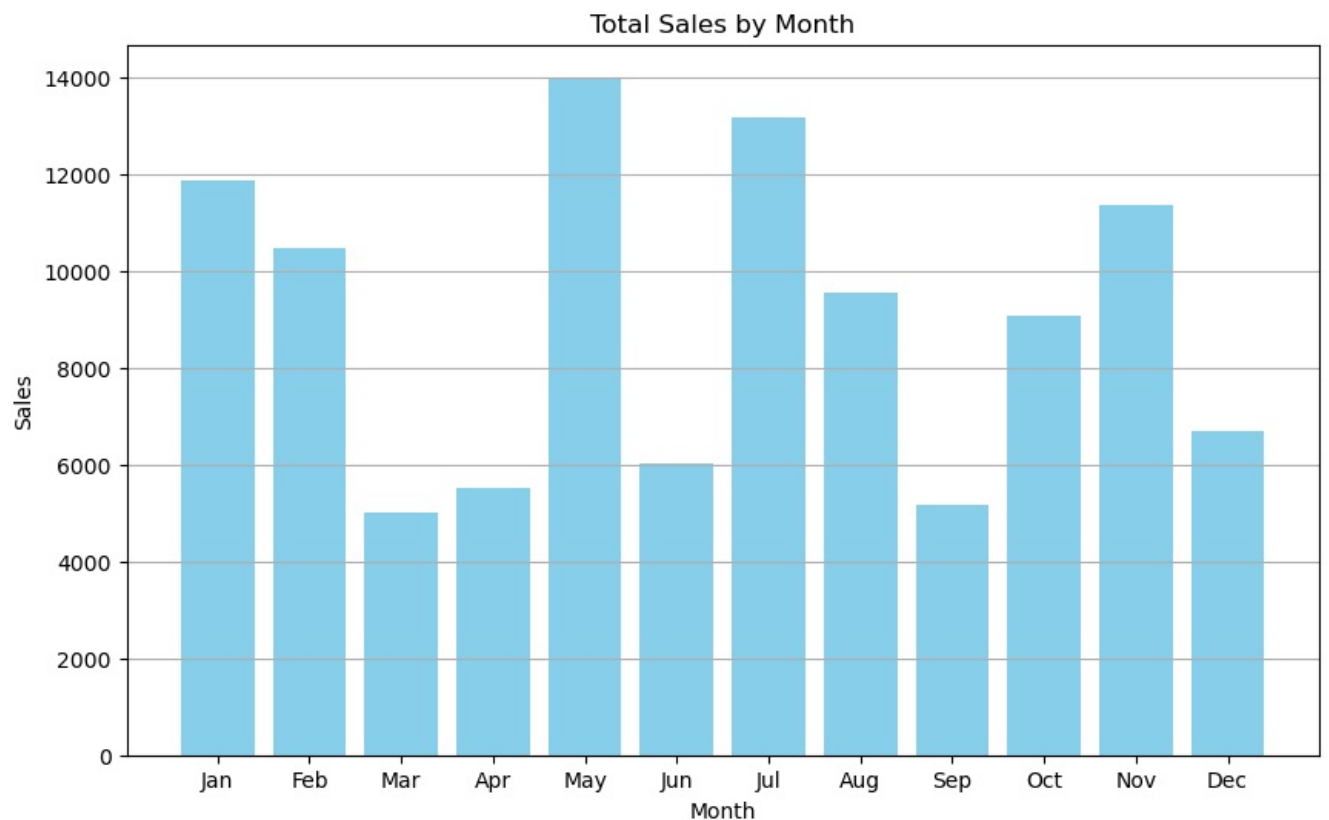
```
In [2]: import seaborn as sns
import matplotlib.pyplot as plt
# Data for hours studied and test scores
hours_studied = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
test_scores = [93, 57, 61, 54, 51, 53, 87, 81, 83, 85]

# Create a scatter plot
plt.figure(figsize=(8, 6))
sns.scatterplot(x=hours_studied, y=test_scores)
plt.title('Hours Studied vs. Test Scores')
plt.xlabel('Hours Studied')
plt.ylabel('Test Scores')
plt.grid(True)
plt.show()
```



```
In [3]: # Data for monthly sales
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]
sales = [11860, 10480, 4997, 5523, 13965, 6011, 13158, 9533, 5158, 9058, 11346, 6675]

# Create a bar chart
plt.figure(figsize=(10, 6))
plt.bar(months, sales, color='skyblue')
plt.title('Total Sales by Month')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.grid(axis='y')
plt.show()
```



In []: